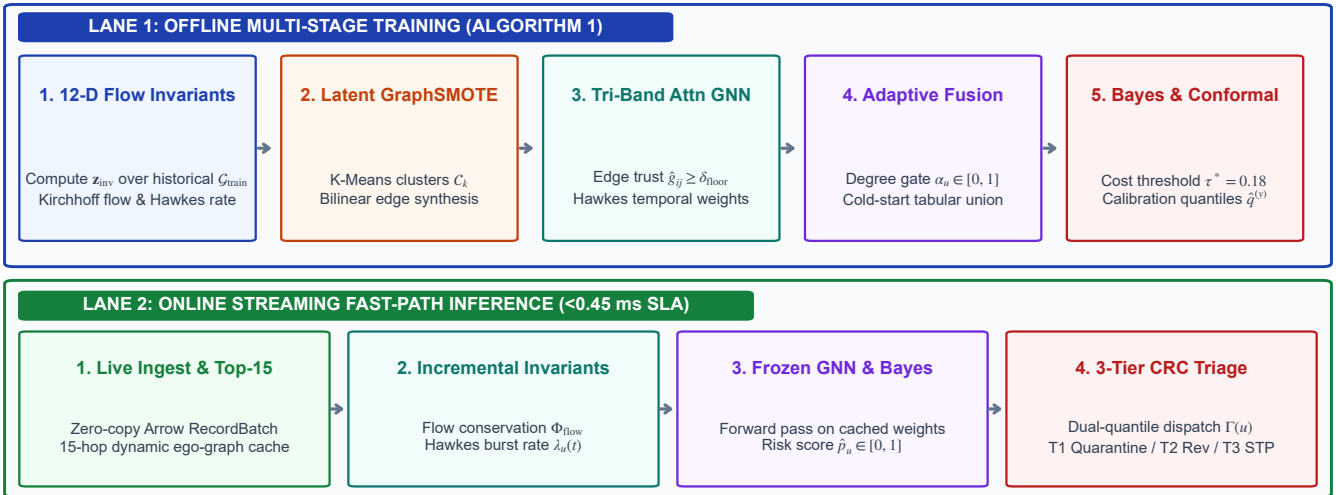

LANE 1: OFFLINE MULTI-STAGE TRAINING (ALGORITHM 1)
1. 12-D Flow Invariants
Compute $\mathbf{z}_{\text{inv}}$ over historical $\mathcal{G}_{\text{train}}$
Kirchhoff flow & Hawkes rate
2. Latent GraphSMOTE
K-Means clusters $\mathcal{C}_k$
Bilinear edge synthesis
3. Tri-Band Attn GNN
Edge trust $\hat{s}_{ij} \geq \delta_{\text{floor}}$
Hawkes temporal weights
4. Adaptive Fusion
Degree gate $\alpha_u \in [0, 1]$
Cold-start tabular union
5. Bayes & Conformal
Cost threshold $\tau^* = 0.18$
Calibration quantiles $\hat{q}^{(v)}$
LANE 2: ONLINE STREAMING FAST-PATH INFERENCE (<0.45 ms SLA)
1. Live Ingest & Top-15
Zero-copy Arrow RecordBatch
15-hop dynamic ego-graph cache
2. Incremental Invariants
Flow conservation $\Phi_{\text{flow}}$
Hawkes burst rate $\lambda_u(t)$
3. Frozen GNN & Bayes
Forward pass on cached weights
Risk score $\hat{p}_u \in [0, 1]$
4. 3-Tier CRC Triage
Dual-quantile dispatch $\Gamma(u)$
T1 Quarantine / T2 Rev / T3 STP